

Physics 7B — Discussion 2 Solutions
Spring 2026

1. Thermal Expansion: Sphere in a Cubic Case

The side length of the plexiglas case and the radius of the gold sphere at temperature T are

$$L(T) = L (1 + \alpha_{\text{plexi}} \Delta T), \quad (1)$$

$$R(T) = R (1 + \alpha_{\text{Au}} \Delta T), \quad (2)$$

where $\Delta T = T - T_0$. The sphere fits as long as

$$L(T) \geq 2R(T). \quad (3)$$

This gives

$$L - 2R = (2R\alpha_{\text{Au}} - L\alpha_{\text{plexi}}) \Delta T, \quad (4)$$

so the maximum allowable temperature is

$$T_m = T_0 + \frac{L - 2R}{2R\alpha_{\text{Au}} - L\alpha_{\text{plexi}}}. \quad (5)$$

2. Wine Bottle

(a)

The fractional volume change of the wine is

$$\frac{\Delta V}{V} = \beta_{\text{wine}} \Delta T = (4.2 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}) (-5 \text{ } ^\circ\text{C}) = -2.1 \times 10^{-3}. \quad (6)$$

Using $V_{\text{wine}} = 750 \text{ mL} = 750 \text{ cm}^3$ gives

$$|\Delta V| = (2.1 \times 10^{-3})(750 \text{ cm}^3) = 1.575 \text{ cm}^3. \quad (7)$$

The initial neck volume is

$$V_{\text{neck}} = \pi(d/2)^2 H = \pi(1 \text{ cm})^2(1.5 \text{ cm}) = 4.712 \text{ cm}^3, \quad (8)$$

so

$$\frac{|\Delta V|}{V_{\text{neck}}} \approx \frac{1.575}{4.712} \approx 0.334. \quad (9)$$

(b)

Since the cross-sectional area is fixed,

$$H' = H \left(1 + \frac{|\Delta V|}{V_{\text{neck}}} \right) = 1.5 \text{ cm} \times 1.334 = 2.00 \text{ cm}. \quad (10)$$

(c)

Treat the trapped air above the wine as ideal:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}. \quad (11)$$

Here $P_1 = 1 \text{ atm}$, $T_1 = 293 \text{ K}$, $T_2 = 288 \text{ K}$, and $V_2/V_1 = 1.334$, hence

$$P_2 = P_1 \frac{T_2}{T_1} \frac{V_1}{V_2} = (1 \text{ atm}) \frac{288}{293} \frac{1}{1.334} = 0.737 \text{ atm}. \quad (12)$$

The force holding the cork in is $F = \Delta P A$ with $A = \pi(d/2)^2 = \pi(1 \text{ cm})^2$:

$$F = (1 - 0.737) \text{ atm} \times 101325 \text{ Pa/atm} \times \pi(1 \times 10^{-2} \text{ m})^2 = 8.37 \text{ N}. \quad (13)$$

(d)

From part (a), a 5°C change corresponds to $|\Delta V|/V_{\text{neck}} = 0.334$. Assuming linear scaling, to expand enough to fill the neck requires $(\Delta V/V_{\text{neck}}) = 1$:

$$\Delta T_{\text{max}} \approx \frac{1}{0.334} (5^\circ\text{C}) = 14.97^\circ\text{C}, \quad (14)$$

so $T_m \approx 20^\circ\text{C} + 14.97^\circ\text{C} \approx 35.0^\circ\text{C}$.

3. Hole and Rivet

(a)

$$R(T) = R_0 [1 + \alpha_s(T - T_a)]. \quad (15)$$

Since $T < T_a$ and $\alpha_s > 0$, the radius decreases: the hole shrinks upon cooling.

(b)

The rivet radius is initially $r_0 = 1.01R_0$ and expands/contracts as

$$r(T) = r_0 (1 + \alpha_r(T - T_a)). \quad (16)$$

To fit exactly into the hole at T_a , require $r(T_r) = R_0$:

$$R_0 = 1.01R_0 (1 + \alpha_r(T_r - T_a)) \quad \Rightarrow \quad 1 = 1.01 (1 + \alpha_r \Delta T), \quad (17)$$

where $\Delta T = T_r - T_a$. Thus

$$\alpha_r \Delta T = \frac{1}{1.01} - 1 = -\frac{0.01}{1.01} = -\frac{1}{101}, \quad (18)$$

so

$$T_r = T_a - \frac{1}{101 \alpha_r}. \quad (19)$$

4. Floating Sphere

At temperature T , the sphere floats half-submerged. Buoyancy balance gives

$$m_s g = \rho_L(T) V_{\text{sub}} g, \quad (20)$$

and with $V_{\text{sub}} = \frac{1}{2} V_s$ this implies

$$\rho_s(T) = \frac{1}{2} \rho_L(T). \quad (21)$$

At $T + \Delta T$, the sphere has neutral buoyancy:

$$\rho_s(T + \Delta T) = \rho_L(T + \Delta T). \quad (22)$$

With fixed mass, density changes only due to volume expansion:

$$\rho_s(T + \Delta T) = \frac{\rho_s(T)}{1 + \beta_s \Delta T}, \quad \rho_L(T + \Delta T) = \frac{\rho_L(T)}{1 + \beta_L \Delta T}. \quad (23)$$

Substitute $\rho_s(T) = \frac{1}{2} \rho_L(T)$:

$$\frac{\frac{1}{2} \rho_L(T)}{1 + \beta_s \Delta T} = \frac{\rho_L(T)}{1 + \beta_L \Delta T} \quad \Rightarrow \quad 1 + \beta_L \Delta T = 2(1 + \beta_s \Delta T). \quad (24)$$

Therefore,

$$\boxed{\beta_L = \frac{1 + 2\beta_s \Delta T}{\Delta T} = \frac{1}{\Delta T} + 2\beta_s.} \quad (25)$$

5. Ice and Water

Energy bookkeeping (heat exchange within the mixture):

- Energy to warm the ice (0.200 kg) from -10°C to 0°C :

$$Q_1 = m_i c_{\text{ice}}(10 \text{ K}) = (0.200 \text{ kg})(2.100 \text{ kJ kg}^{-1} \text{ K}^{-1})(10 \text{ K}) = 4.20 \text{ kJ}. \quad (26)$$

- Energy available by cooling the water (1.00 kg) from 15°C to 0°C :

$$Q_w = m_w c_{\text{water}}(15 \text{ K}) = (1.00 \text{ kg})(4.186 \text{ kJ kg}^{-1} \text{ K}^{-1})(15 \text{ K}) = 62.79 \text{ kJ}. \quad (27)$$

After paying Q_1 to warm the ice, the remaining energy goes into melting an amount m' of ice:

$$Q_1 + L_f m' = Q_w \quad \Rightarrow \quad m' = \frac{Q_w - Q_1}{L_f} = \frac{62.79 \text{ kJ} - 4.20 \text{ kJ}}{333 \text{ kJ kg}^{-1}} = 0.176 \text{ kg}. \quad (28)$$

Since $0.176 \text{ kg} < 0.200 \text{ kg}$, not all the ice melts, so the final equilibrium temperature is

$$T_f = 0^\circ\text{C}.$$

Final masses:

$$m_{\text{ice,final}} = 0.200 \text{ kg} - 0.176 \text{ kg} = 0.024 \text{ kg}, \quad m_{\text{water,final}} = 1.00 \text{ kg} + 0.176 \text{ kg} = 1.176 \text{ kg}. \quad (29)$$

Entropy (qualitative)

A closed system does *not* imply constant entropy: S stays constant only if the process is reversible. Here heat flows from warmer water to colder ice through a finite temperature difference and some ice melts. Both effects produce entropy, so the total entropy of the (closed) ice+water system *increases*.

6. First Law of Thermodynamics

We use the first law in the form

$$\Delta E = -W + Q, \quad (30)$$

where W is the work *done by the gas*.

(a)

For an ideal gas, equipartition gives

$$C_V = \frac{f}{2}R. \quad (31)$$

With $C_V = 3R$,

$$\frac{f}{2}R = 3R \quad \Rightarrow \quad f = 6. \quad (32)$$

A monatomic ideal gas has $f = 3$, so this cannot be monatomic.

(b)

For one mole of an ideal gas, the internal energy is

$$E(T) = nC_V T = (1)(3R)T = 3RT, \quad (33)$$

so

$$\Delta E = 3R\Delta T. \quad (34)$$

(i) **Isothermal.** By definition, $\Delta T = 0$, hence

$$\Delta E = 3R\Delta T = 0. \quad (35)$$

Then the first law gives

$$0 = -W + Q \quad \Rightarrow \quad W = Q. \quad (36)$$

(ii) **Isochoric.** Here V is constant, so

$$W = \int P dV = 0. \quad (37)$$

Thus the first law becomes

$$\Delta E = Q. \quad (38)$$

Using $\Delta E = 3R\Delta T$ gives

$$3R\Delta T = Q \quad \Rightarrow \quad \Delta T = \frac{Q}{3R}, \quad \Delta E = Q, \quad W = 0. \quad (39)$$

(iii) **Isobaric.** Here $P = P_0$ is constant. The work done by the gas is

$$W = \int_{V_1}^{V_2} P_0 dV = P_0(V_2 - V_1). \quad (40)$$

Using the ideal gas law $P_0V = RT$ (one mole),

$$V_1 = \frac{RT_1}{P_0}, \quad V_2 = \frac{RT_2}{P_0} \quad \Rightarrow \quad V_2 - V_1 = \frac{R(T_2 - T_1)}{P_0} = \frac{R\Delta T}{P_0}, \quad (41)$$

so

$$W = P_0 \left(\frac{R\Delta T}{P_0} \right) = R\Delta T. \quad (42)$$

Now plug $\Delta E = 3R\Delta T$ and $W = R\Delta T$ into the first law:

$$3R\Delta T = -R\Delta T + Q \quad \Rightarrow \quad 4R\Delta T = Q. \quad (43)$$

Therefore,

$$\Delta T = \frac{Q}{4R}, \quad W = R\Delta T = \frac{1}{4}Q, \quad \Delta E = 3R\Delta T = \frac{3}{4}Q. \quad (44)$$